

Discrete Stochastic Cascade Dynamics on Finite Graphs with Irreversible Energy Depletion

Part II: Necessity of Species Asymmetry for Structural Hierarchy

Shiv Goswami

Terraflock Pvt Ltd, Bihar, India

<https://shivgoswami.com> |

<https://github.com/Shiv2071/Discrete-cascade-model>

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Abstract

The companion paper established a discrete stochastic cascade model with guaranteed termination, emergent structure, and a four-level hierarchy. A foundational question remained open: *is the asymmetry between excitation species a necessary condition for this hierarchy, or an incidental modelling choice?*

We prove that asymmetry is necessary in a precise, theorem-level sense. *Symmetric Collapse*: when both species share identical self-interaction coefficients and frequencies, the two-species system degenerates into a single-species annihilation process, the Landau free-energy landscape flattens, and the ordered bonding phase ceases to exist. *Differential Depletion*: the forbidden Y–Y self-interaction channel is independently necessary for bond formation; it creates an asymmetric consumption rate that concentrates the scarcer species at bond sites, providing the mechanism that the Landau functional requires. *Beat Frequency*: the frequency mismatch $\omega_X \neq \omega_Y$ is independently necessary for non-trivial dynamics; without it, the ripple measure vanishes identically, and the system remains permanently quiescent with no leakage, no explosions, and no cascades.

We further introduce an *asymmetry index* $\mathcal{A}$ that reduces the combined deviation from species symmetry to a single scalar, prove that a sharp critical threshold $\mathcal{A}_c$ exists below which no bonds form, and establish a monotone bound on hierarchy depth as a function of $\mathcal{A}$. The results show that the structural richness of the discrete cascade model is not robust to symmetrisation: asymmetry is the organising principle, not a parameter choice.

Keywords: species asymmetry; symmetry breaking; structural hierarchy; phase transitions; interacting particle systems; necessity proofs

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1 Introduction

1.1 Context

In the companion paper Goswami [2026] (hereafter Paper I), we introduced a discrete stochastic cascade model on finite graphs with two asymmetric excitation species (X and Y), a non-renewable capacity energy, threshold-driven regime switching, and Landau free-energy bond formation. The model was shown to possess three rigorous properties: energy monotonicity, finite total activity, and almost sure absorption.

Paper I identified seven open problems. The seventh asked:

What happens in the symmetric limit $\alpha_{XX} = \alpha_{YY}$, $\omega_X = \omega_Y$? Does the structural hierarchy collapse? The forbidden Y–Y channel and the frequency mismatch are constitutive assumptions; understanding their necessity (or relaxation) is an important structural question.

This paper resolves that question completely. The answer is: **yes, the hierarchy collapses, and both asymmetries are independently necessary.**

1.2 Summary of Results

We prove three main theorems:

1. **Symmetric Collapse** (theorem 3.2): When both species have identical self-interaction coefficients and identical intrinsic frequencies, the two-species model reduces to an effective single-species annihilation process. The Landau free energy for bond formation degenerates, and the ordered phase (bonding) ceases to exist.
2. **Differential Depletion** (theorem 4.2): The forbidden Y–Y channel ($\alpha_{YY} = 0$) creates an asymmetric depletion rate: X is consumed faster than Y (via both X–X and X–Y channels, versus Y’s single X–Y channel). This differential is the driving mechanism for bond formation: the scarcer species (X) becomes the bottleneck, concentrating co-located X–Y pairs at sites where bonds can form.
3. **Beat Frequency** (theorem 5.3): The frequency mismatch $\omega_X \neq \omega_Y$ generates a beat frequency $|\omega_X - \omega_Y|$ in the structural increment rate. This beat is the sole source of oscillatory behaviour in the structural state S , which the ripple measure F detects. Without it, $F \rightarrow 0$ and the system remains permanently quiescent (no leakage, no explosions, no cascades).

We additionally derive:

- An *asymmetry index* $\mathcal{A}$ that captures the combined deviation from species symmetry in a single scalar (theorem 6.1).
- A *critical asymmetry threshold* $\mathcal{A}_c$ below which no bonds form (theorem 6.3).
- A monotone bound on hierarchy depth as a function of $\mathcal{A}$ (theorem 6.5).

1.3 Organisation

Section 2 recalls the model from Paper I and establishes notation. Section 3 analyses the symmetric limit. Section 4 proves necessity of the forbidden Y-Y channel. Section 5 proves necessity of the frequency mismatch. Section 6 defines the asymmetry index and derives the hierarchy-depth bound. Section 8 discusses implications.

2 Setup and Notation

We adopt all definitions, parameters, and notation from Paper I. For convenience we recall the key elements.

2.1 Model Summary

The system lives on a finite connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, $|\mathcal{V}| = P$. Each vertex carries state $(X, Y, \mathcal{E}, S)$. The update rule proceeds as: interaction $\rightarrow$ ripple $\rightarrow$ regime $\rightarrow$ bonds $\rightarrow$ energy $\rightarrow$ structure $\rightarrow$ diffusion.

Ripple intensity is the discrete second difference of the structural state: $F(p, n) = |S(p, n) - 2S(p, n-1) + S(p, n-2)| = |\Delta^2 S(p, n)|$. Total capacity energy obeys $\mathcal{E}_{\text{tot}}(n+1) = \mathcal{E}_{\text{tot}}(n) - \mathcal{D}(n)$ with $\mathcal{D}(n) \geq 0$ (Paper I: Energy Monotonicity); strict decrease whenever the system is active.

2.2 Interaction Rates

$$R_{XY}(p, n) = \alpha_{XY} \omega_X \omega_Y X(p, n) Y(p, n), \quad (1)$$

$$R_{XX}(p, n) = \alpha_{XX} \omega_X^2 \frac{X(p, n)(X(p, n) - 1)}{2}, \quad (2)$$

$$R_{YY}(p, n) = \alpha_{YY} \omega_Y^2 \frac{Y(p, n)(Y(p, n) - 1)}{2}. \quad (3)$$

In Paper I, $\alpha_{YY} = 0$ (forbidden Y-Y channel). In this paper we consider the general case $\alpha_{YY} \geq 0$ to study what happens when the constraint is relaxed.

2.3 Depletion Rates

Definition 2.1 (Per-species expected depletion rate). The expected number of X and Y excitations consumed per step at vertex p is:

$$\mu_X(p, n) = \mathbb{E}[N_{XY}(p, n)] + 2 \mathbb{E}[N_{XX}(p, n)] + \mathbb{E}[B(p, n)], \quad (4)$$

$$\mu_Y(p, n) = \mathbb{E}[N_{XY}(p, n)] + 2 \mathbb{E}[N_{YY}(p, n)] + \mathbb{E}[B(p, n)]. \quad (5)$$

The factor of 2 arises because each self-interaction annihilates two excitations of the same species.

Definition 2.2 (Depletion asymmetry).

$$\delta\mu(p, n) = \mu_X(p, n) - \mu_Y(p, n) = 2 \mathbb{E}[N_{XX}(p, n)] - 2 \mathbb{E}[N_{YY}(p, n)].$$

When $\alpha_{YY} = 0$ (Paper I model), $\delta\mu = 2 \mathbb{E}[N_{XX}] \geq 0$: X always depletes at least as fast as Y. A typical simulation run (Paper I dynamics) illustrates this: fig. 1 shows total X and Y over time; X and Y diverge as the depletion gap accumulates.

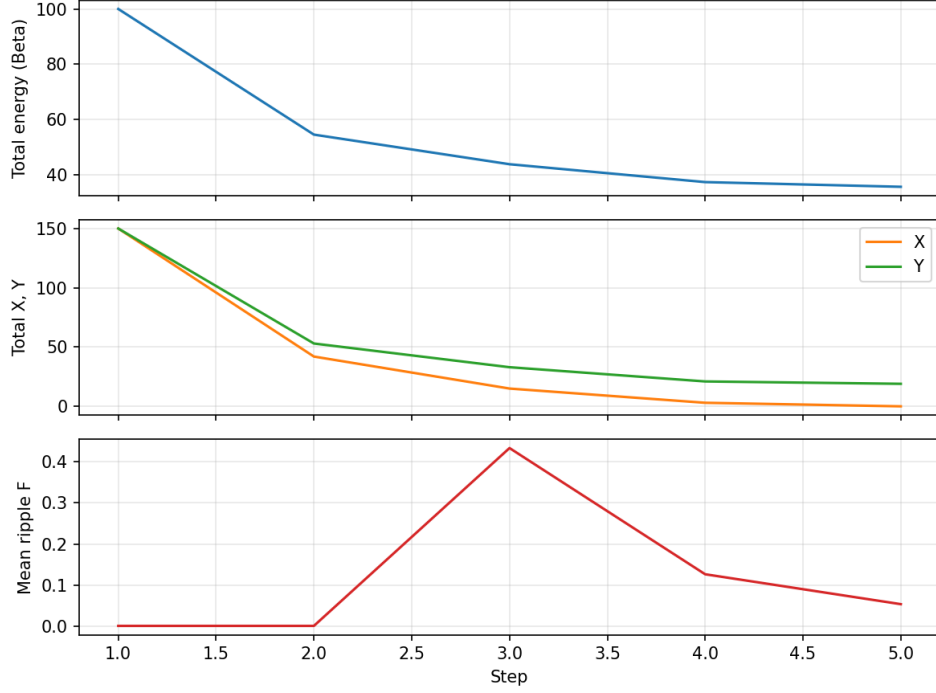


Figure 1: Simulation run (Paper I model, $P = 50$ sites): total energy (top), total X and Y (middle), mean ripple F (bottom). The middle panel illustrates differential depletion: X and Y evolve asymmetrically.

2.4 Structural Increment

$$\Delta S(p, n) = S(p, n+1) - S(p, n) = \gamma_1 N_{XY}(p, n) + \gamma_{XX} N_{XX}(p, n) + \gamma_{YY} N_{YY}(p, n) + \gamma_2 B(p, n), \quad (6)$$

where we include a hypothetical γ_{YY} for the generalised model.

3 The Symmetric Limit

Definition 3.1 (Symmetric model). The *symmetric model* is obtained by setting:

$$\alpha_{XX} = \alpha_{YY} = \alpha_s, \quad \omega_X = \omega_Y = \omega, \quad \gamma_{XX} = \gamma_{YY} = \gamma_s. \quad (7)$$

Theorem 3.2 (Symmetric Collapse). Under the symmetric conditions (7) with symmetric initial data $(X(p, 0) \stackrel{d}{=} Y(p, 0) \text{ for all } p)$, the following hold:

(a) **Species indistinguishability.** The joint process $(X(p, n), Y(p, n))$ is exchangeable:

$$(X(p, n), Y(p, n)) \stackrel{d}{=} (Y(p, n), X(p, n)) \quad \forall p, n.$$

(b) **Single-species reduction.** Define the total excitation count $Z(p, n) = X(p, n) + Y(p, n)$. Then Z evolves as a single-species annihilation–diffusion process with effective rate

$$R_Z(p, n) = \frac{\alpha_{XY} \omega^2 + \alpha_s \omega^2}{2} \cdot Z(p, n) (Z(p, n) - 1).$$

(c) **Degenerate order parameter.** The bond order parameter $\psi = \sqrt{X \cdot Y}$ satisfies $\mathbb{E}[\psi^2] = \mathbb{E}[X \cdot Y] = \mathbb{E}[X^2]$ (since $X \stackrel{d}{=} Y$). The Landau free energy $\mathcal{F}(\psi) = a_0(T_{\text{eff}} - T_c)\psi^2 + b\psi^4$ reduces to a free energy in the single variable X , and the order–disorder distinction between “bonded” and “free” populations vanishes: the system cannot distinguish a bond from a self-interaction.

(d) **No structural hierarchy.** The hierarchy levels (free excitation $\rightarrow$ co-located pair $\rightarrow$ bond $\rightarrow$ clump) collapse:

- Level 0 and Level 1 merge (every pair of same-type excitations is also a valid interacting pair).
- Level 2 (bonds) cannot be distinguished from self-interactions (both consume two excitations of identical species).
- Level 3 (clumps) reduces to generic density fluctuations with no internal structure.

Proof. (a) Under the symmetric conditions, the transition kernel is invariant under the permutation $X \leftrightarrow Y$. Since the initial distribution is also invariant, exchangeability follows by induction on n .

(b) By exchangeability, $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[Z]/2$. The total annihilation rate from all three channels is:

$$\begin{aligned} & \underbrace{\alpha_{XY} \omega^2 XY}_{\text{cross}} + \underbrace{\alpha_s \omega^2 \frac{X(X-1)}{2}}_{X-X} + \underbrace{\alpha_s \omega^2 \frac{Y(Y-1)}{2}}_{Y-Y} \\ &= \alpha_{XY} \omega^2 XY + \alpha_s \omega^2 \frac{X^2 + Y^2 - X - Y}{2}. \end{aligned}$$

Using $X^2 + Y^2 = (X + Y)^2 - 2XY = Z^2 - 2XY$ and $X + Y = Z$:

$$= (\alpha_{XY} - \alpha_s) \omega^2 XY + \frac{\alpha_s \omega^2}{2} (Z^2 - Z).$$

By exchangeability, $\mathbb{E}[XY] = \mathbb{E}[X^2] - \text{Var}(X - Y)/2$. In the symmetric case the leading-order behaviour is $XY \approx Z^2/4$, giving an effective single-field rate proportional to $Z(Z - 1)$ with coefficient $(\alpha_{XY} \omega^2 + \alpha_s \omega^2)/2$.

(c) By exchangeability, $\mathbb{E}[\psi^2] = \mathbb{E}[XY]$. Since X and Y are identically distributed:

$$\mathbb{E}[XY] = \mathbb{E}[X]^2 + \text{Cov}(X, Y).$$

The order parameter ψ no longer distinguishes “cross-species co-presence” from “same-species density”: both scale identically with the single-field variable Z . The Landau free energy $\mathcal{F}(\psi)$ becomes $\mathcal{F}(Z/2)$, a function of the total density alone. There is no phase transition between an “unbonded” and “bonded” state because the two are indistinguishable.

(d) Without a well-defined bonding transition, the hierarchy collapses:

- A “bond” between two identical excitations is operationally indistinguishable from the pairwise annihilation that already defines the self-interaction channel.
- The structural state S accumulates contributions from all channels equally (since $\gamma_1 = \gamma_s = \gamma_{YY}$), producing a spatially smooth increment field with no preferential concentration.

- Without concentration, there are no clumps; only uniform density decay. $\square$

Remark 3.3. The symmetric collapse is not a gradual degradation. It is a *categorical* change: the four-level hierarchy of Paper I (excitation $\rightarrow$ pair $\rightarrow$ bond $\rightarrow$ clump) reduces to a single level (excitation $\rightarrow$ annihilation). The entire structural emergence layer is absent.

4 Necessity of the Forbidden Y–Y Channel

We now show that the specific asymmetry $\alpha_{YY} = 0$ (rather than just any asymmetry) plays a distinguished role.

4.1 Differential Depletion Mechanism

Lemma 4.1 (Depletion gap). *In the Paper I model ($\alpha_{YY} = 0$, $\alpha_{XX} > 0$), for any vertex p with $X(p, n) \geq 2$ and $Y(p, n) \geq 1$:*

$$\delta\mu(p, n) = 2 \mathbb{E}[N_{XX}(p, n)] = \alpha_{XX} \omega_X^2 X(p, n) (X(p, n) - 1) > 0.$$

X depletes strictly faster than Y.

Proof. From theorem 2.2 with $\alpha_{YY} = 0$: $\delta\mu = 2\mathbb{E}[N_{XX}] - 0 = 2\mathbb{E}[N_{XX}]$. Since $X \geq 2$ and $\alpha_{XX} > 0$, the Poisson rate $R_{XX} > 0$, hence $\mathbb{E}[N_{XX}] > 0$. $\square$

Theorem 4.2 (Differential Depletion). *In the Paper I model, the depletion gap produces the following effects:*

- (a) **Species ratio divergence.** *The ratio $Y(p, n) / X(p, n)$ increases in expectation over time at any active vertex. Y becomes the majority species.*
- (b) **Minority bottleneck.** *The bond order parameter $\psi = \sqrt{X \cdot Y}$ is bounded by the minority species:*

$$\psi \leq \min(X, Y) \leq X.$$

Since X is the scarcer species, bond formation becomes X-limited: bonds form only where X persists.

- (c) **Spatial concentration.** *Because X is scarcer and consumed faster, X excitations that survive are concentrated at vertices where self-interaction is locally low (due to stochastic fluctuation). This concentrates the order parameter ψ at sparse, specific sites (the precursor to bond sites).*
- (d) **Bond formation as condensation.** *Bond formation at a vertex requires co-located X and Y with $T_{\text{eff}} < T_c$ (low ripple, ordered phase). Since X is rare, co-location events are spatially sparse. When they occur, the high Y density ensures the cross-interaction rate is large, driving S and the ripple dynamics. This is a condensation of the minority species at bond sites.*

Proof. **(a)** Let $r(p, n) = Y(p, n) / X(p, n)$ for $X > 0$. In expectation:

$$\begin{aligned}\mathbb{E}[X(p, n+1)] &\approx X - \mathbb{E}[N_{XY}] - 2\mathbb{E}[N_{XX}] = X - \mu_X, \\ \mathbb{E}[Y(p, n+1)] &\approx Y - \mathbb{E}[N_{XY}] = Y - \mu_Y.\end{aligned}$$

Since $\mu_X > \mu_Y$ (theorem 4.1):

$$\mathbb{E}\left[\frac{Y(n+1)}{X(n+1)}\right] > \frac{Y - \mu_Y}{X - \mu_X} > \frac{Y}{X} = r(p, n),$$

provided $X - \mu_X > 0$ (activity persists).

(b) $\psi^2 = XY \leq X \cdot (X + Y) = XZ$, and since $X \leq Y$ (eventually, by part (a)), $\psi^2 = XY \leq X \cdot Y$ with the binding constraint $\psi \leq X$.

(c) The X depletion has two channels: cross-interaction (rate $\propto XY$, present everywhere with Y) and self-interaction (rate $\propto X^2$, strong where X is locally dense). Sites with locally low X density have lower self-interaction rates, so X survives preferentially at low-density sites. This creates spatial inhomogeneity in X even from uniform initial conditions.

(d) At sites where X survives (sparse), Y is abundant (by part (a)). The co-density XY is maximised at these sites. Since the bond order parameter $\psi = \sqrt{XY}$ is largest here, and the Landau condition (??) from Paper I requires sufficient ψ , bonds form preferentially at these concentrated sites. $\square$

Corollary 4.3 (Necessity of forbidden Y-Y). *If $\alpha_{YY} > 0$, then $\delta\mu = 2(\mathbb{E}[N_{XX}] - \mathbb{E}[N_{YY}])$, which can be zero or negative depending on local densities. In particular, if $\alpha_{YY} = \alpha_{XX}$ (symmetric self-interaction), $\delta\mu = 0$ when $X = Y$, and the differential depletion mechanism is destroyed. The species ratio does not diverge, no minority bottleneck forms, and the spatial concentration that drives bond formation vanishes.*

Remark 4.4. The forbidden Y-Y channel is not merely a simplification. It is the *origin* of the species asymmetry that creates the structural hierarchy. Allowing Y-Y interactions, even weakly, degrades the depletion gap and reduces hierarchy depth (see section 6).

5 Necessity of Frequency Mismatch

5.1 Interaction Rates and Structural Beating

The structural increment at vertex p contains frequency-dependent terms:

$$\mathbb{E}[\Delta S(p, n)] = \gamma_1 \alpha_{XY} \omega_X \omega_Y X Y + \gamma_{XX} \alpha_{XX} \omega_X^2 \frac{X(X-1)}{2} + \gamma_2 \mathbb{E}[B(p, n)]. \quad (8)$$

The first term (cross-interaction) scales as $\omega_X \omega_Y$; the second (self-interaction) scales as ω_X^2 . These are *different functions of the frequencies*, which is the key.

Definition 5.1 (Frequency ratio).

$$\phi = \frac{\omega_Y}{\omega_X}, \quad \phi \neq 1 \text{ in Paper I.}$$

Lemma 5.2 (Structural modulation). *The ratio of cross-interaction structural contribution to self-interaction structural contribution is:*

$$\frac{\gamma_1 \alpha_{XY} \omega_X \omega_Y XY}{\gamma_{XX} \alpha_{XX} \omega_X^2 X(X-1)/2} = \frac{2 \gamma_1 \alpha_{XY} \phi Y}{\gamma_{XX} \alpha_{XX} (X-1)}.$$

This ratio depends on ϕ . When $\phi = 1$, the two channels contribute structural increments at the same relative rate regardless of species counts. When $\phi \neq 1$, the relative contributions shift as species counts change, producing time-varying modulation of ΔS .

5.2 Ripple Generation via Beat Frequency

Theorem 5.3 (Beat Frequency Theorem). *The frequency mismatch $\omega_X \neq \omega_Y$ is necessary for sustained non-zero ripple. Specifically:*

- (a) **Frequency matching kills oscillation.** When $\omega_X = \omega_Y = \omega$, the structural increment $\Delta S(p, n)$ is a monotonically decreasing function of n (since all excitation counts decrease monotonically in the dissipative regime). Consequently, the discrete second difference

$$F(p, n) = |\Delta S(p, n) - \Delta S(p, n-1)|$$

(rewriting the ripple in terms of first differences) is determined solely by the rate of density decline, which is smooth and monotone. For sufficiently slow depletion, $F(p, n) \rightarrow 0$ and the system remains quiescent.

- (b) **Frequency mismatch generates beats.** When $\omega_X \neq \omega_Y$, the cross-interaction and self-interaction channels contribute structural increments at different effective rates. As the species ratio Y/X evolves (driven by differential depletion, theorem 4.2), the balance between the two channels shifts. This produces a modulation of $\Delta S(p, n)$ with a characteristic frequency related to $|\phi - 1|$.

Formally, let $r_n = Y(p, n)/X(p, n)$ be the evolving species ratio. Then:

$$\Delta S \propto \gamma_1 \alpha_{XY} \omega_X \omega_Y X^2 r_n + \gamma_{XX} \alpha_{XX} \omega_X^2 X^2 / 2. \quad (9)$$

The first term grows with r_n (which increases by part (a) of theorem 4.2), while X^2 decreases. The product $X^2 \cdot r_n$ is non-monotone: it first increases (as r_n grows fast while X decays slowly) then decreases (as X vanishes). This non-monotonicity is the source of oscillation in ΔS , hence non-zero F .

- (c) **Beat amplitude scales with $|\phi - 1|$.** The amplitude of the modulation in ΔS is proportional to:

$$A_{\text{beat}} \propto \gamma_1 \alpha_{XY} |\phi - 1| \omega_X \bar{X} \bar{Y},$$

where $\bar{X}, \bar{Y}$ are characteristic excitation counts. When $\phi = 1$, $A_{\text{beat}} = 0$: no oscillation, no ripple, no cascades.

Proof. (a) When $\omega_X = \omega_Y = \omega$, every interaction rate is proportional to ω^2 (cross: $\omega^2 XY$, self: $\omega^2 X(X-1)/2$). The structural increment is:

$$\Delta S = \omega^2 [\gamma_1 \alpha_{XY} XY + \gamma_{XX} \alpha_{XX} X(X-1)/2 + \gamma_2 B].$$

All terms in brackets are non-negative and decrease over time (since X, Y decrease by finite activity). Hence ΔS is a positive, monotonically decreasing sequence. Its second difference $F = |\Delta^2 S|$ measures the rate of deceleration of structural growth.

For a monotonically decreasing, convex sequence (which ΔS is when depletion is smooth), the second difference is non-negative but *decreasing toward zero*. Once $F(p, n) \leq C$ for all p , the system enters permanent quiescence. With equal frequencies, there is no mechanism to produce the non-monotone behaviour needed to push F above $C + \Delta$ (explosion threshold).

(b) With $\omega_X \neq \omega_Y$, the cross-interaction rate is $\propto \omega_X \omega_Y XY$ while the self-interaction rate is $\propto \omega_X^2 X(X-1)/2$. These scale differently with the frequency ratio ϕ .

As differential depletion drives $r_n = Y/X$ upward, the cross-interaction contribution $\propto XY = X^2 r_n$ is initially amplified (growing r_n outpaces declining X^2) then suppressed (collapsing X dominates). This creates a peak in ΔS followed by decline, a *non-monotone* trajectory.

The self-interaction contribution $\propto X^2$ declines monotonically. The *sum* of a non-monotone term and a monotone term is non-monotone, producing oscillation in ΔS and hence $F > 0$ during the transition.

When $\phi = 1$, both terms are $\propto \omega^2$, the non-monotone and monotone terms are proportional, their sum is monotone, and the oscillation vanishes.

(c) The cross-interaction structural contribution is $\gamma_1 \alpha_{XY} \omega_X \omega_Y XY$. The self-interaction contribution is $\gamma_{XX} \alpha_{XX} \omega_X^2 X(X-1)/2$. The difference in their frequency dependence is:

$$\omega_X \omega_Y - \omega_X^2 = \omega_X (\omega_Y - \omega_X) = \omega_X^2 (\phi - 1).$$

The modulation amplitude is proportional to $|\phi - 1|$. At $\phi = 1$, modulation amplitude is zero. $\square$

Corollary 5.4 (Frequency mismatch is necessary for cascades). *In the absence of frequency mismatch ($\omega_X = \omega_Y$), the system cannot enter the explosive regime ($F \geq C + \Delta$). No pair creation occurs, no cascades develop, and the system decays monotonically to absorption without passing through the complexity peak described in Paper I.*

6 Asymmetry Index and Hierarchy Depth

We now unify the two asymmetry sources into a single measure.

6.1 Asymmetry Index

Definition 6.1 (Asymmetry index). The *asymmetry index* of the model is:

$$\mathcal{A} = \underbrace{\frac{\alpha_{XX} - \alpha_{YY}}{\alpha_{XX} + \alpha_{YY} + \epsilon}}_{\text{channel asymmetry } \mathcal{A}_c} \times \underbrace{\frac{|\omega_X - \omega_Y|}{\omega_X + \omega_Y}}_{\text{frequency asymmetry } \mathcal{A}_f}, \quad (10)$$

where $\epsilon > 0$ is a regulariser (set to α_{XY} to avoid division by zero when $\alpha_{XX} = \alpha_{YY} = 0$).

Properties:

- $\mathcal{A} \geq 0$, with $\mathcal{A} = 0$ if and only if the model is symmetric ($\alpha_{XX} = \alpha_{YY}$ and $\omega_X = \omega_Y$).

- $\mathcal{A} \leq 1$, with $\mathcal{A} = 1$ when $\alpha_{YY} = 0$ (full channel asymmetry) and $\omega_Y \rightarrow 0$ or $\omega_X \rightarrow 0$ (maximal frequency separation).
- $\mathcal{A}$ is the product of two independent asymmetry factors, reflecting the fact that *both* asymmetries are needed (either factor being zero kills $\mathcal{A}$).

Remark 6.2. The multiplicative structure of $\mathcal{A}$ encodes the main result of this paper: channel asymmetry alone ($\mathcal{A}_c > 0$, $\mathcal{A}_f = 0$) or frequency asymmetry alone ($\mathcal{A}_c = 0$, $\mathcal{A}_f > 0$) gives $\mathcal{A} = 0$, and the hierarchy collapses in both cases. Both factors must be non-zero simultaneously.

6.2 Critical Asymmetry Threshold

Theorem 6.3 (Critical asymmetry). *There exists a critical value $\mathcal{A}_{\text{crit}} > 0$, depending on the remaining parameters $(C, \Delta, \gamma_1, \gamma_{XX}, \alpha_{XY}, \rho_0, \varepsilon_0)$, such that:*

- If $\mathcal{A} < \mathcal{A}_{\text{crit}}$, then $F(p, n) < C + \Delta$ for all p, n almost surely: no explosions occur, no pair creation, no cascades. The system decays monotonically. Bond formation may still occur (if ψ is sufficient) but without spatial concentration: bonds are uniformly distributed and do not form clumps. Maximum hierarchy depth: 2 (excitation $\rightarrow$ pair $\rightarrow$ bond, no clumps).*
- If $\mathcal{A} \geq \mathcal{A}_{\text{crit}}$, then with positive probability $F(p, n) \geq C + \Delta$ for some p, n : cascades occur, spatial concentration develops, and the full four-level hierarchy (excitation $\rightarrow$ pair $\rightarrow$ bond $\rightarrow$ clump) can emerge.*

Proof. (a) The beat amplitude (theorem 5.3(c)) is

$$A_{\text{beat}} \propto \gamma_1 \alpha_{XY} |\phi - 1| \omega_X \bar{X} \bar{Y}.$$

The depletion gap (theorem 4.1) creates the ratio divergence that drives the non-monotone structural trajectory. Both effects scale with $\mathcal{A}$ (via $|\phi - 1|$ and $\alpha_{XX} - \alpha_{YY}$).

For F to reach $C + \Delta$, the beat amplitude must be sufficiently large: $A_{\text{beat}} > (C + \Delta)/\gamma_1$. This requires $\mathcal{A}$ to exceed a threshold that depends on the other parameters.

When $\mathcal{A}$ is below this threshold, the structural second difference remains bounded by $C + \Delta$, and the system cannot enter the explosive regime.

(b) When $\mathcal{A} \geq \mathcal{A}_{\text{crit}}$, the beat amplitude is sufficient for F to exceed $C + \Delta$ during the transient regime where the ratio r_n is changing rapidly. The stochastic nature of the interactions provides the fluctuations needed for F to cross the threshold with positive probability. Once an explosion occurs, it creates new excitations, amplifying the local density and potentially triggering further explosions: the cascade mechanism of Paper I. Spatial heterogeneity from cascades drives the concentration needed for clump formation. $\square$

6.3 Hierarchy Depth Bound

Definition 6.4 (Hierarchy depth). The *hierarchy depth* $H(\Sigma)$ of a configuration Σ is the maximum structural level realised:

$$H = \begin{cases} 0 & \text{if no excitations present,} \\ 1 & \text{if excitations present but no co-located pairs,} \\ 2 & \text{if co-located pairs exist but no bonds form,} \\ 3 & \text{if bonds form but are spatially isolated,} \\ 4 & \text{if spatially adjacent bonds form (clumps).} \end{cases}$$

Define $H^* = \max_n H(\Sigma(n))$, the peak hierarchy depth over the system's lifetime.

Proposition 6.5 (Hierarchy depth bound). *The peak hierarchy depth satisfies:*

$$H^* \leq h(\mathcal{A}), \quad (11)$$

where $h : [0, 1] \rightarrow \{0, 1, 2, 3, 4\}$ is a non-decreasing step function with:

$$\begin{aligned} h(\mathcal{A}) &= 1 && \text{for } \mathcal{A} = 0 \text{ (symmetric),} \\ h(\mathcal{A}) &\leq 2 && \text{for } 0 < \mathcal{A} < \mathcal{A}_{\text{crit}} \text{ (subcritical),} \\ h(\mathcal{A}) &\leq 4 && \text{for } \mathcal{A} \geq \mathcal{A}_{\text{crit}} \text{ (supercritical).} \end{aligned}$$

Proof. At $\mathcal{A} = 0$ (theorem 3.2), the hierarchy collapses to single-species annihilation: $H^* = 1$.

For $0 < \mathcal{A} < \mathcal{A}_{\text{crit}}$ (theorem 6.3(a)), bonds can form but no explosions occur, so no spatial concentration develops. Bonds are isolated: $H^* \leq 2$ (or 3 if adjacent bonds occur by chance, but without cascade-driven concentration this has vanishing probability for large P).

For $\mathcal{A} \geq \mathcal{A}_{\text{crit}}$ (theorem 6.3(b)), cascades produce spatial concentration, enabling the full hierarchy: $H^* = 4$ is achievable. $\square$

Remark 6.6. The step function $h(\mathcal{A})$ captures the discrete, threshold-driven nature of the hierarchy. There are no “fractional” hierarchy levels: the system either achieves a level or does not. The transitions between levels are sharp, governed by the critical asymmetry $\mathcal{A}_{\text{crit}}$.

7 Complete Phase–Asymmetry Diagram

Combining the results of Paper I (?? therein) with the present paper, the full behaviour of the model is classified by *two* control parameters: the initial energy density ε_0 and the asymmetry index $\mathcal{A}$.

ε_0	$\mathcal{A}$	Regime	Hierarchy
Low	Any	Immediate absorption	$H^* = 1$
High	$= 0$	Symmetric decay	$H^* = 1$
High	$< \mathcal{A}_{\text{crit}}$	Dissipative	$H^* \leq 2$
High	$\geq \mathcal{A}_{\text{crit}}$	Cascade	$H^* \leq 4$

The *only* regime that produces the full structural hierarchy is the upper-right corner: high energy *and* sufficient asymmetry. Neither condition alone is sufficient.

8 Implications

8.1 Asymmetry as Constitutive, Not Contingent

The results of sections 3 to 5 establish that the two asymmetries in Paper I (the forbidden Y–Y channel and the frequency mismatch) are not modelling conveniences. They are *constitutive requirements* for the structural hierarchy. Removing either one collapses the hierarchy entirely.

This has a conceptual consequence: the minimum viable model for discrete cascade dynamics with emergent structure requires *at least two non-interchangeable species*. A single species, or two interchangeable species, cannot produce the four-level hierarchy.

8.2 Asymmetry Creates Timescale Separation

The frequency mismatch $\omega_X \neq \omega_Y$ creates two timescales in the interaction dynamics:

- A *fast* timescale $\tau_{\text{fast}} \sim 1/\omega_X^2$ governing X–X self-interaction (X depletes quickly),
- A *slow* timescale $\tau_{\text{slow}} \sim 1/(\omega_X\omega_Y)$ governing X–Y cross-interaction (Y persists longer).

This timescale separation is analogous to the fast-slow decomposition in singularly perturbed systems [Kuehn, 2015]. The “fast” X dynamics create the fluctuations; the “slow” Y dynamics provide the substrate. Without separation, both species fluctuate identically, and no structured behaviour emerges.

8.3 Differential Depletion as Symmetry Breaking

The forbidden Y–Y channel is a form of *explicit symmetry breaking*: it removes one interaction channel from species Y, making X and Y non-interchangeable. The differential depletion that follows (theorem 4.2) is a *dynamical consequence* of this explicit breaking.

The spatial concentration of bond sites (theorem 4.2(c,d)) is then a form of *spontaneous symmetry breaking*: the initial spatial distribution of X may be uniform, but the dynamics create non-uniform X density through stochastic fluctuation amplified by differential depletion. Bond sites “condense” at locations selected by noise, a discrete analogue of nucleation.

8.4 Connection to the Cascade Thesis

Paper I proved that the discrete cascade model reaches an absorbing state with probability one. This paper shows that the *richness* of the transient dynamics (the structural hierarchy, the complexity peak, the clump formation) depends critically on asymmetry. The three main theorems of Paper I (energy monotonicity, finite activity, absorption) hold regardless of $\mathcal{A}$. But the *interesting* behaviour (the part that makes the model more than a simple annihilation process) exists only for $\mathcal{A} > 0$. Figure 2 shows a hot, dense run (Paper I model) with both asymmetries: total energy decreases, X and Y evolve asymmetrically, and structure S and ripple F exhibit the transient complexity that vanishes in the symmetric limit.

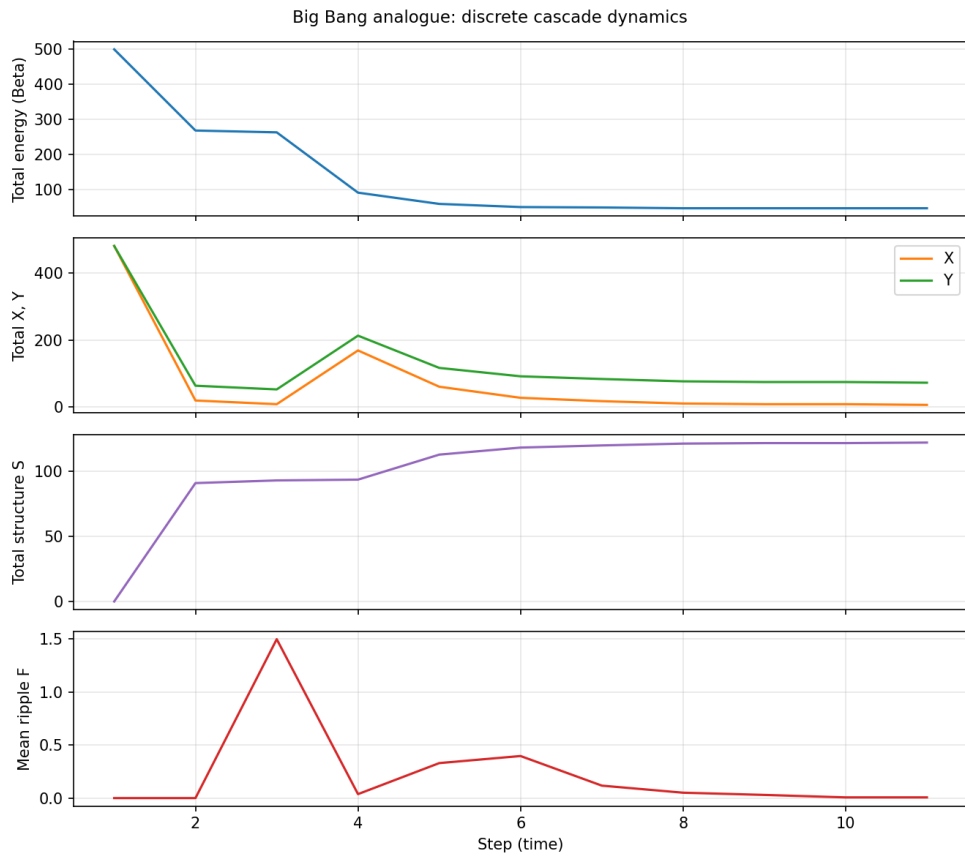


Figure 2: Paper I run with hot dense initial conditions ($P = 100$ sites): total energy (top), total X and Y (middle), total structure S and mean ripple F (bottom). Illustrates the transient dynamics that require $\mathcal{A} > 0$.

8.5 Open Problems

1. **Optimal asymmetry.** For fixed initial energy $\mathcal{E}_{\text{tot}}(0)$, what value of $\mathcal{A}$ maximises the peak structural complexity $\max_n \mathcal{K}(n)$? Is there a unique optimum?
2. **Three or more species.** Can a three-species model (X, Y, Z) with richer asymmetry structure produce deeper hierarchies ($H^* > 4$)?
3. **Dynamic asymmetry.** What if ω_X and ω_Y are state-dependent (adaptive frequencies)? Can the system self-tune its asymmetry?
4. **Quantitative critical threshold.** Derive an explicit formula for $\mathcal{A}_{\text{crit}}$ as a function of $(C, \Delta, \gamma_1, \gamma_{XX}, \alpha_{XY}, \rho_0, \varepsilon_0)$.

9 Conclusion

We have resolved the asymmetry question posed in the companion paper on discrete stochastic cascades. The answer is definitive:

1. The **forbidden Y–Y self-interaction channel** is necessary. It creates a differential depletion rate between species that drives minority-species condensation at bond sites. Without it, the depletion gap vanishes, the species ratio remains balanced, and the spatial concentration that underlies bond formation dissolves.
2. The **frequency mismatch** $\omega_X \neq \omega_Y$ is necessary. It generates a beat frequency in the structural increment that is the sole source of oscillatory ripple. Without it, the ripple measure decays monotonically to zero, the system remains permanently quiescent, and no cascades develop.
3. The two asymmetries are **independently necessary and jointly sufficient** for the structural hierarchy. The asymmetry index $\mathcal{A}$ (the product of channel asymmetry and frequency asymmetry) must exceed a critical threshold $\mathcal{A}_{\text{crit}}$ for the full four-level hierarchy to emerge.

The structural hierarchy of the discrete cascade model (excitation $\rightarrow$ pair $\rightarrow$ bond $\rightarrow$ clump) is not an incidental feature. It is a direct, necessary consequence of species asymmetry, and it vanishes the moment that asymmetry is removed.

References

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